Lecture 3

January 15, 2026

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Embedded Options Change Interest Rate Risk

- Callable bonds
- Puttable bonds
- Mortgage-like structures
- Convertible bonds

Callable Bonds

- A callable bond embeds a call option owned by the issuer, allowing early redemption at a fixed call price.
- Callability introduces negative convexity and reinvestment risk for investors.
- Common features: call dates, call price schedule, call protection period.

Puttable Bonds

- A puttable bond embeds a put option owned by the investor, allowing early redemption of the bond back to the issuer at a fixed put price.
- Putability introduces positive convexity and downside price protection for investors.
- Common features: put dates, put price schedule, put protection period.

Duration for Bond w/Embedded Options

- For such bonds, duration is path-dependent and state-dependent.
- Because of the presence of embedded options, cashflows provided by the bonds are no longer known.
- This makes such instruments difficult to analyze in terms of the duration measures we have used up until this point.

Modified Duration Breaks Down

- Modified duration assumes:
  - Fixed cash flows
  - No option exercise
  - No change in expected maturity
- Bonds with embedded options violate these assumptions:
  - Cash flows change as rates move
  - Expected maturity shortens or extends
  - Price response becomes asymmetric
- As a result:
  - Modified duration misstates true rate sensitivity
  - Errors are largest when options are near the money

Effective Duration

- Effective duration is designed for any bond with rate-dependent cash flows, i.e., bonds with embedded options.
- Price sensitivity to interest rate changes after accounting for how cash flows change when rates move.
- That makes it appropriate for:
  - Callable bonds
  - Puttable bonds
  - MBS / ABS
  - Any instrument with optionality

Effective Duration: General Mechanics

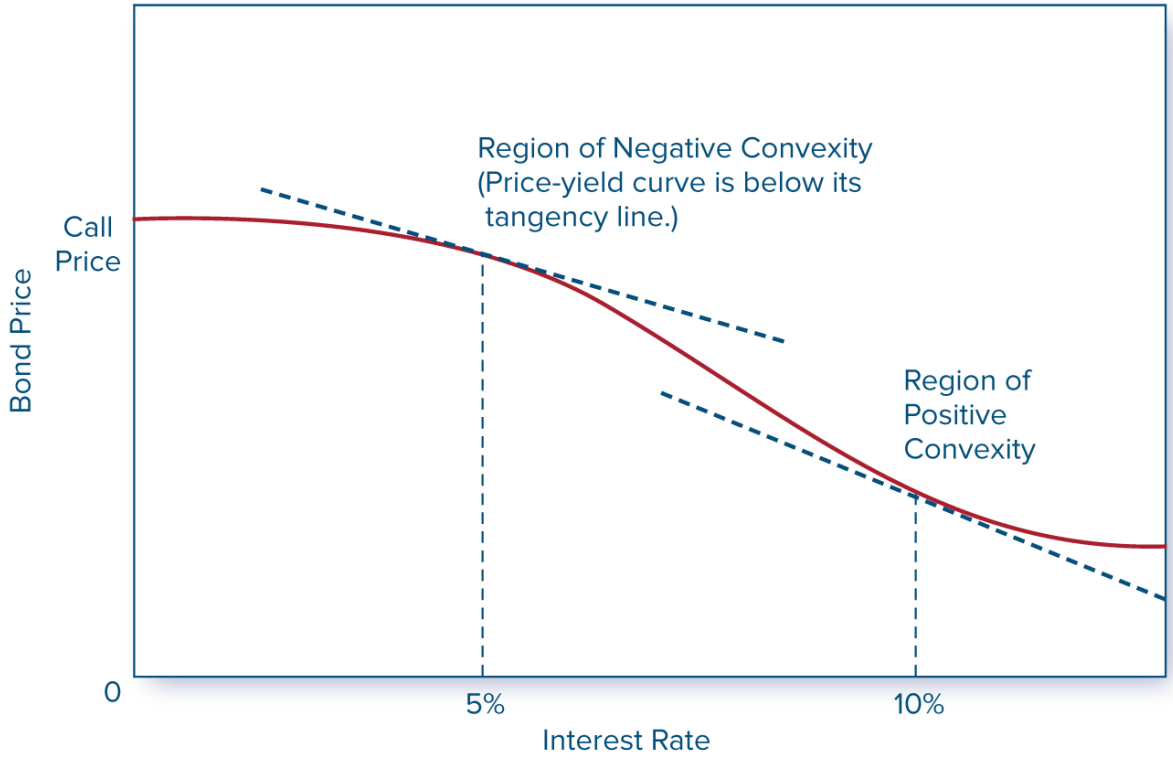
- Effective duration measures interest rate sensitivity by:
  - Holding credit spreads and option assumptions constant
  - Shocking the yield curve up and down
  - Repricing the bond under each scenario
- Effective Duration captures:
  - Changes in discount rates
  - Changes in expected cash flows
  - Changes in option exercise behavior
- This produces a state-dependent duration.

Callable Bonds: Effective Duration Intuition

- Callable bonds embed an issuer call option.
  - When interest rates fall:
    - Call probability rises
    - Expected maturity shortens
    - Price appreciation is capped
    - Effective duration declines
  - When interest rates rise:
    - Call risk disappears
    - Expected maturity extends
    - Effective duration increases
  - Result:
    - Asymmetric rate sensitivity (complicates hedging and portfolio construction)
    - Negative convexity

Negative Convexity for a Callable Bond

- Unlike straight bonds, callable bonds exhibit negative convexity.
- Price appreciation is capped when rates fall (call risk).
- Price declines fully when rates rise (extension risk).



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Puttable Bonds: Effective Duration Intuition

- Puttable bonds embed an investor put option.
  - When interest rates rise:
    - Put probability rises
    - Expected maturity shortens
    - Downside price risk is limited
    - Effective duration declines
  - When interest rates fall:
    - Put option moves out of the money
    - Bond behaves more like a straight bond
    - Effective duration increases
  - Result:
    - Downside protection
    - Positive convexity

Calculating Effective Duration

- Effective duration measures the percentage price change of a bond for a small change in interest rates, accounting for changes in cash flows.
- Formula: $D_{eff} = -(\Delta P / P) / \Delta r$

  or equivalently: $D_{eff} = [\, P(-\Delta r) - P(+\Delta r) \,] / [\, 2 \times P_0 \times \Delta r \,]$
- Where:
  P_0 = initial bond price
  $\Delta P / P$ = percentage price change
  Δr = change in yield (in decimal form)

Effective Duration Callable Bond Example

- Suppose a callable bond with a call price of $1,040 is selling today for $995.
  - If the yield curve shifts up by 0.25%, the bond price falls to $970
  - If the yield curve shifts down by 0.25%, the bond price rises to $1,015
- To compute effective duration, we evaluate the bond price under symmetric interest-rate shocks.
- Step 1: Change in interest rates
  - $\Delta r = \text{Assumed increase in rates} - \text{Assumed decrease in rates}$
  - $\Delta r = 0.25\% - (-0.25\%) = 0.50\% = 0.005$

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Effective Duration Callable Bond Example

- Step 2: Change in bond price
  - ΔP=Price at +0.25% rates−Price at −0.25% rates
  - ΔP=970−1,015=−45

- Step 3: Calculate effective duration
  - Effective Duration = $-(\Delta P / P) / \Delta r$
    = -(−45/995)/ 0.005 =9.05 years
- The bond's price changes by approximately 9.05%for a 1 percentage point parallel shift in interest rates around the current yield environment

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Let’s drop this in a portfolio with Plain Vanillas

- Convert Effective Duration to DV01
  - Price = 99.5 (using 100 par convention)
  - $D_{eff} = 9.05$
  - DV01 per \$100 = $D \times P \times 0.0001$
  - $DV01 = 9.05 \times 99.5 \times 0.0001 \approx 0.090$
- The bond’s price changes by about 9 cents for a 1 bp move, per \$100.

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In A Portfolio

Par ($MM)	Instrument	Price (per $100)	Duration Type	Duration
4.0	UST 5yr	99.40	Modified	4.10
3.0	UST 10yr	98.90	Modified	7.60
2.0	Callable Corp	99.50	**Effective**	9.05
—	**Portfolio Total**	—	—	—

- For a +1 bp parallel increase in rates the portfolio value falls by?
- What are the steps to get the answer? (hint: we just did it for the Callable Bond)

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Mixed Portfolio

Par ($MM)	Instrument	Price (per $100)	Duration Type	Duration	DV01 per $100 ($/bp)	Position DV01 ($/bp)
4.0	UST 5yr	99.40	Modified	4.10	0.041	1,640
3.0	UST 10yr	98.90	Modified	7.60	0.075	2,250
2.0	Callable Corp	99.50	**Effective**	9.05	0.090	1,800
—	**Portfolio Total**	—	—	—	—	**5,690**

- For a +1 bp parallel increase in rates the portfolio value falls by ≈ $5,690
- What are we assuming here?

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Mixed Portfolio (con.)

Par ($MM)	Instrument	Price (per $100)	Duration Type	Duration	DV01 per $100 ($/bp)	Position DV01 ($/bp)
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2.0	Callable Corp	99.50	**Effective**	9.05	0.090	1,800
—	**Portfolio Total**	—	—	—	—	**5,690**

- What are we assuming here?
  - Same curve
  - Same parallel shift
  - Small move (local linear approximation)

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Mortgage-Backed Structures

- Mortgage-backed securities embed a prepayment option held by the borrower.
- Actually, the biggest market for which call provisions are important.
- Prepayment is economically similar to a call option on the bond.
- Cash flows are explicitly rate-dependent and path-dependent.
- As a result, modified duration is inappropriate for MBS.
- Effective duration is the correct measure of interest-rate sensitivity.

Effective Duration Behavior of MBS

Rates Fall

- Refinancing incentives increase
- Prepayment speeds accelerate
- Expected maturity shortens
- Effective duration declines

Rates Rise

- Refinancing incentives disappear
- Prepayments slow
- Expected maturity lengthens
- Effective duration increases

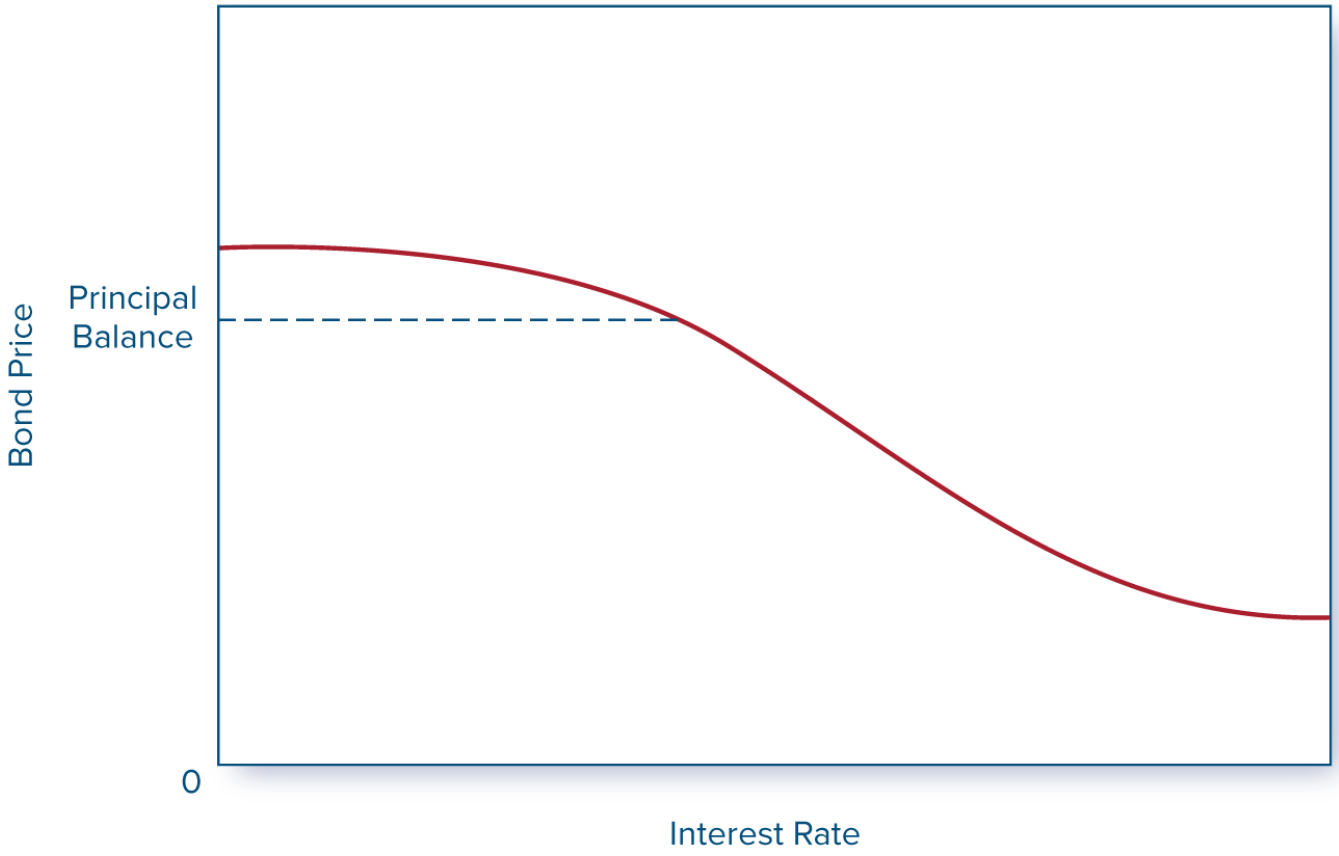
MBS vs Callable and Puttable Bonds

- Structural Comparison
  - Callable bonds: discrete issuer call dates
  - Puttable bonds: discrete investor put dates
  - MBS: continuous, stochastic borrower prepayment
- Implications:
  - MBS exhibit strong negative convexity.
  - Effective duration can change rapidly.
  - OAS[1] and effective duration must be interpreted jointly.

1:Option-Adjusted Spread (OAS) is the constant spread added to the risk-free yield curve that equates the model-implied price of a bond with embedded options to its market price. OAS is discussed in subsequent slides.

Price-yield curve for an MBS

- Why might an MBS trade above the principal amount?



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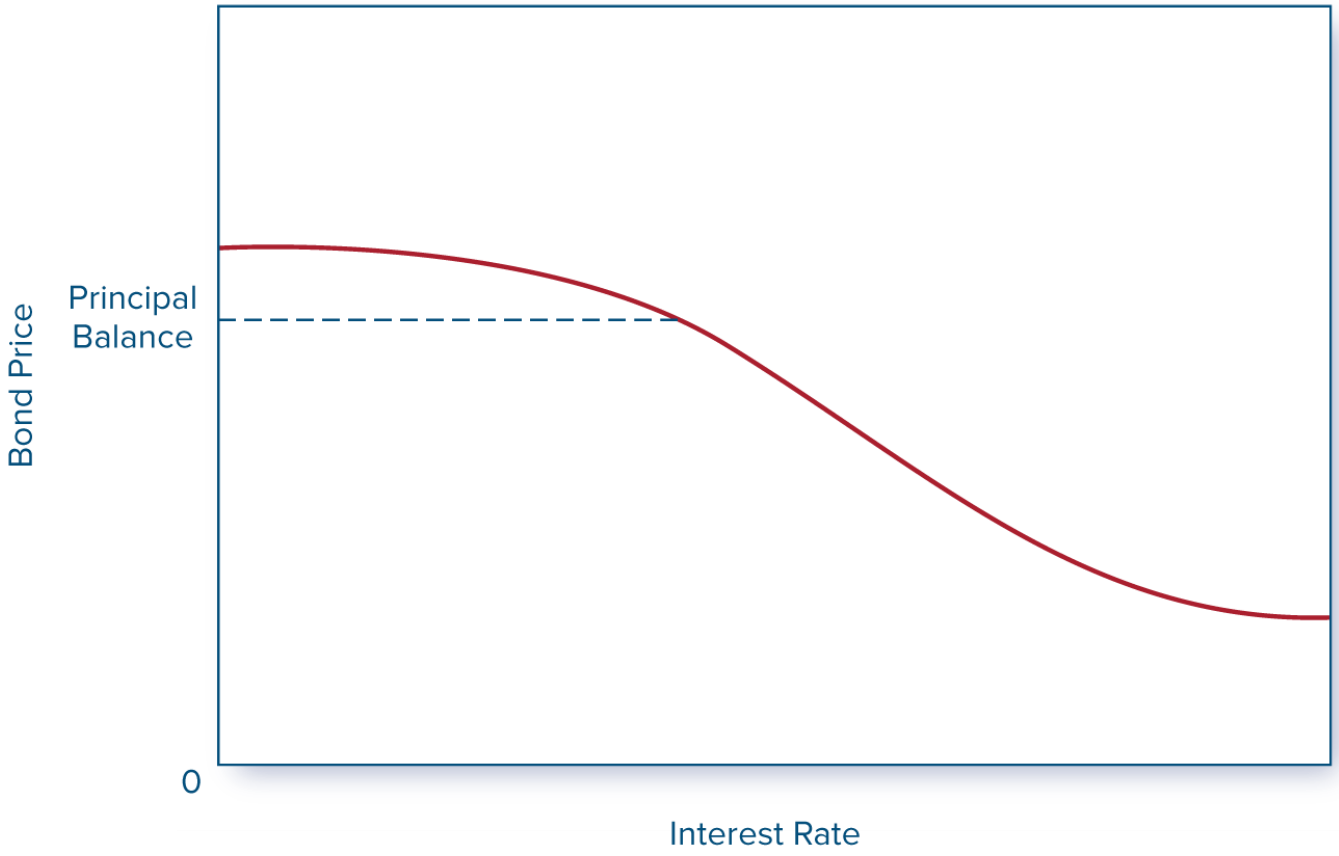
Price-yield curve for an MBS

- While an MBS security exhibits negative convexity at low rates, its implicit call price is not a firm ceiling on its value.
- Homeowners:
  - Do not refinance their loans as soon as rates drop
  - May not want to incur costs or hassles related to refinance
  - Lack sophistication



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Mortgage-Backed Derivatives

- Mortgage-backed derivatives redistribute the cash flows and risks of mortgage-backed securities through structured designs.
- Straight (Pass-Through) MBS
  - Pool residential mortgage loans and pass principal and interest directly to investors
  - Can be agency MBS (GNMA, FNMA, FHLMC) or non-agency MBS
  - Investors bear prepayment, extension, and interest-rate risk
  - Agency MBS serve as the primary collateral for agency CMOs
- Agency CMOs
  - Backed by agency MBS (GNMA, FNMA, FHLMC)
  - Minimal credit risk due to explicit or implicit government guarantees
  - Tranching is used primarily to redistribute interest rate and prepayment risk
- Non-Agency CMOs
  - Backed directly by pools of private-label mortgage loans or RMBS
  - Tranche mortgage cash flows into senior and subordinated bonds
  - Redistribute credit, prepayment, and interest-rate risk
  - Cash flows come directly from underlying mortgages

Mortgage-Backed Derivatives

- Mortgage-Backed CDOs
  - CDOs whose collateral consists of mortgage-related debt instruments
  - May include RMBS, CMO tranches, or other mortgage securities
  - Designed to redistribute credit and correlation risk
  - May involve re-securitization, but this is not universal
- Synthetic Mortgage Derivatives
  - Transfer mortgage credit risk using derivatives rather than physical collateral
  - Designed for credit exposure rather than interest rate management

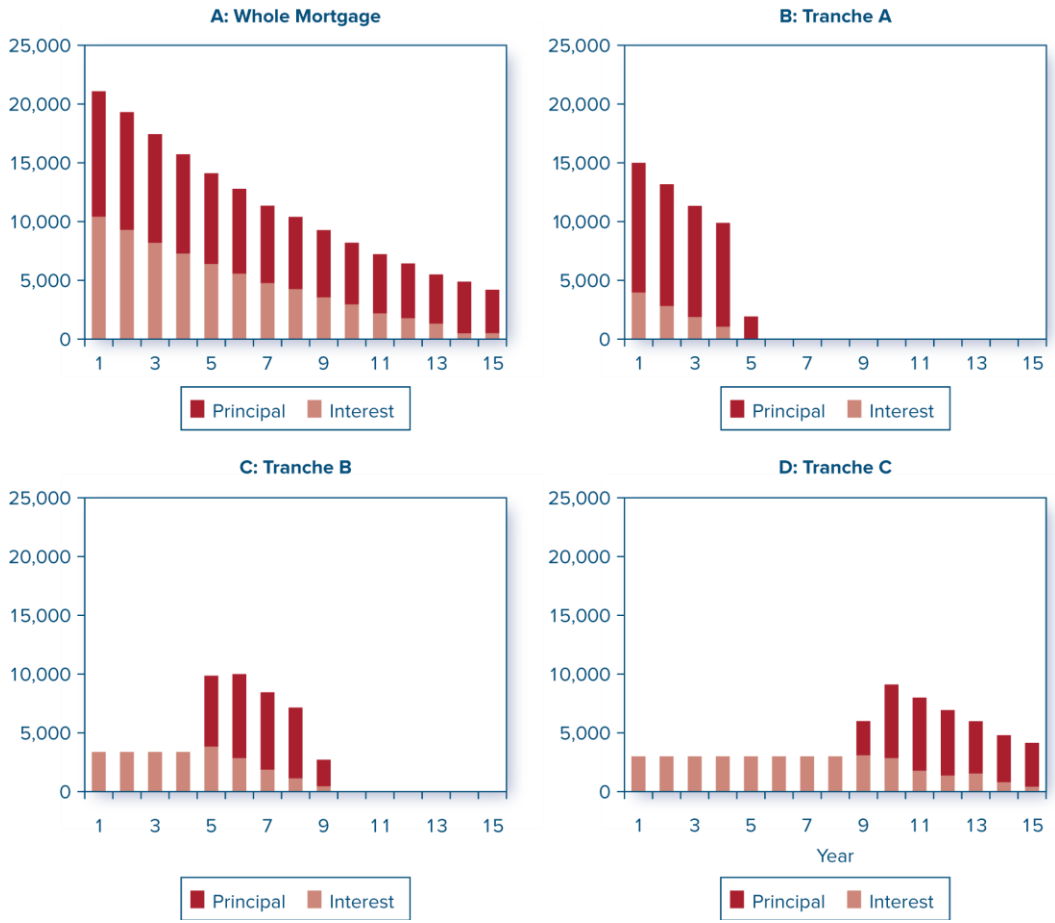
Agency CMOs and Interest Rate Risk Management

- Agency CMOs are structured securities backed by agency pass-through MBS, designed specifically to redistribute interest rate and prepayment risk.
  - Credit risk is largely removed through agency guarantees, isolating rate-driven cash-flow risk
  - Tranching reallocates prepayment timing across investors, reshaping effective duration and convexity
  - Structures such as sequential-pay, PAC, and support tranches manage:
    - Extension risk
    - Contraction risk
    - Duration stability across rate environments
  - As a result, agency CMOs are the primary instruments used to analyze and manage interest rate risk in mortgage markets.

Illustrative CMO

- 3 classes: A, B, and C
- All principal repayments (scheduled & prepayment) channeled first to A, then B, and, finally, C.
- For a standard agency CMO, Tranche A bears more prepayment (reinvestment) risk and less price (interest-rate) risk than longer tranches.
- This is a way to allocate interest rate risk among institutional investors.
- Risks assumed by different classes depend on par value in each class.
- How would risks change in two scenarios below?

Tranche	Scenario A Par	Scenario B Par
A	400	100
B	300	200
C	100	500



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Valuing Callables, Puttables, and MBS

- Callable bonds, puttable bonds, and MBS have rate-dependent cash flows.
  - Cash flows change as interest rates move
  - Expected maturity is not fixed
  - Modified duration and standard DCF assumptions break down
- As a result:
  - Valuation must explicitly account for embedded optionality
  - Interest-rate risk cannot be summarized by a single yield or duration

Price Decomposition: Economic Intuition

- Bonds with embedded options can be decomposed conceptually as:
  - Callable bonds: Straight bond value minus issuer call option
  - Puttable bonds: Straight bond value plus investor put option
  - MBS: Cash flows reduced by borrower prepayment option
- This framework explains:
  - Negative vs positive convexity
  - Asymmetric price behavior as rates change

Why simple discounting fails:

- Traditional valuation assumes:
  - Fixed cash flows
  - A single discount curve
  - No exercise decisions
- For bonds with optionality:
  - Cash flows depend on the path of interest rates
  - Exercise decisions are state-dependent
  - Expected maturity changes dynamically
- Accurate valuation requires modeling both rates and exercise behavior jointly.

State, Path, and Plain-Vanilla Cash Flows

- Plain-Vanilla (Straight) Bonds
  - Cash flows are fixed and contractual
  - Principal and coupon timing do not depend on interest rates
  - Neither state-dependent nor path-dependent
- State-Dependent Cash Flows
  - Cash flows depend on the current level of interest rates
  - Examples:
    - Callable bonds (called when rates are low)
    - Puttable bonds (put when rates are high)
- Path-Dependent Cash Flows
- Cash flows depend on the history of interest rates, not just today's level
- Examples:
  - Mortgage-backed securities (prepayments depend on sustained rate movements)

Term Structure Models and Optimal Exercise

- Option-adjusted valuation uses term structure models to simulate future interest rate paths.
- Common frameworks include:
  - Ho-Lee
  - Black-Derman-Toy
  - Hull-White
- These models:
  - Generate multiple interest rate scenarios
  - Allow optimal call, put, or prepayment decisions at each point in time
  - Produce state-dependent cash flows consistent with economic incentives

Option-Adjusted Spread (OAS)

- In addition to calculating theoretical prices for bonds with embedded options, traders also like to compute what is know as the Option-Adjusted Spread (OAS).
- OAS is used as a relative-value and risk-analysis tool for securities with embedded options.
- This is the constant spread over risk-free term structure provided by the instrument when all options have been taken into account.
- To calculate the OAS for an instrument it is:
  - First priced using zero-coupon Treasury curve
  - Then compared to the price in the market
  - Then iterated to determine parallel shift that causes the model-implied price to be equal to the observed market price

Option-Adjusted Spread (OAS): What It Does

- Option-Adjusted Spread (OAS) is a market-comparable measure for securities with embedded options.
  - Removes the value of embedded options from observed spreads
  - Allows comparison across callable bonds, MBS, and CMOs with different option features
  - Higher OAS reflects greater compensation for credit and liquidity risk, after adjusting for optionality
- How OAS is used
  - Compute effective duration and convexity by holding OAS constant and shocking interest rates
  - Support relative-value analysis and trade-offs between yield and optionality
  - Inform structure selection (e.g., Planned amortization tranches vs support tranches, callable vs non-callable bonds)
- OAS is model-dependent and should be interpreted comparatively, not as a yield.

The Difficulty of Valuing Interest Rate Derivatives

- Interest rate derivatives are more difficult to value than equity and foreign exchange derivatives for the following reasons:
  - The behavior of an individual interest rate is more complicated than that of a stock price or exchange rate.
  - For the valuation of many products, it is necessary to develop a model describing the behavior of the entire zero-coupon curve.
  - The volatilities of different points on the yield curve are different.
  - Interest rates are used for discounting the derivative as well as defining its payoff.
- **Deploying the aforementioned models is outside the scope of this class, but you should be aware these are what are used to price these instruments**

Convertible bonds & Effective Duration

- Convertibles are different
- A convertible bond embeds multiple options:
  - A call option on equity (held by the investor)
  - Often an issuer call
  - Sometimes an investor put
- As a result, price sensitivity is driven by:
  - Interest rates
  - Credit spreads
  - Equity price
  - Volatility
- Effective duration remains the correct measure of rate risk, but its economic relevance varies with the bond's equity sensitivity.

How Effective Duration Behaves for Convertibles

- The usefulness of effective duration depends on where the convertible sits relative to its conversion value.
- Bond-like convertibles (out of the money)
  - Value dominated by fixed-income component
  - Effective duration is meaningful and comparable to callable bonds
- Balanced convertibles (near the money)
  - Both equity and bond components matter
  - Effective duration is reduced and unstable
- Equity-like convertibles (in the money)
  - Value driven primarily by equity
  - Effective duration approaches zero
- Convertibles provide state-dependent interest rate exposure
- Duration hedging is secondary to equity risk management for equity-like convertibles

Valuing Convertible Bonds: Why Standard Methods Fail

- Convertible bonds embed American-style optionality and credit risk.
  - Investors may convert at any time prior to maturity
  - Issuers may have call provisions
  - Cash flows depend on:
    - Interest rates
    - Equity price paths
    - Credit quality and default risk
- As a result:
  - Simple discounted cash flow and closed-form solutions are not sufficient
  - Valuation requires numerical, path-dependent methods

Numerical Valuation Framework

- Convertible bonds are typically valued using binomial or trinomial trees.
  - Trees jointly model:
    - Interest rate evolution
    - Equity price dynamics
  - At each node, the investor optimally chooses between:
    - Holding the bond
    - Converting into equity
    - Accepting a call (if callable)
- This framework:
  - Captures American-style exercise features
  - Allows conversion, call, and maturity decisions to be evaluated consistently

Credit Risk, Default Intensity, and Implementation

- Credit risk is incorporated through default intensity (hazard rate) modeling.
  - Default can occur at any point in time
  - Recovery assumptions determine payoff upon default
  - Credit spreads are reflected via default probabilities rather than discount rates
- In practice:
  - Convertible valuation relies on specialized software platforms
  - Models integrate:
    - Equity dynamics
    - Interest rates
    - Volatility
    - Default risk

Joint Modeling of Equity and Interest Rates

- Convertible bond trees typically combine:
  - An equity price tree (e.g., binomial stock process)
  - An interest rate tree (short-rate model)
- Key features:
  - Equity and interest rate processes may be correlated
  - Default risk is layered in via default intensity at each node
  - Valuation proceeds via backward induction across the joint tree

Portfolio Immunization

- Portfolio immunization is a fixed-income strategy designed to protect the value of a portfolio against interest-rate changes.
- Aims to insure that asset values will be sufficient to meet a known future liability, regardless of small interest-rate movements.
- Balances opposing effects of:
  - Price risk
  - Reinvestment risk
- Typical users are pension funds, insurance companies, and investors with defined future obligations.

Price Risk vs Reinvestment Risk

- Interest rate changes affect bond portfolios through two opposing channels.
- Price Risk
  - Bond prices fall when interest rates rise
  - More significant for longer-duration portfolios
- Reinvestment Risk
  - Cash flows are reinvested at lower rates when interest rates fall
  - More significant for shorter-duration portfolios

How Portfolio Immunization Works

- Core Conditions
  - Present value of assets = present value of liabilities
  - Duration of assets = duration of liabilities
- Implementation
  - Construct a bond portfolio (ladder or barbell) with the required duration
  - Favor higher convexity to improve protection
  - Rebalance periodically as time and interest rates change
- Key Assumptions & Limits
    - Works best for small, parallel yield curve shifts
    - Less effective under large or non-parallel rate changes
    - Does not eliminate credit or liquidity risk

Rebalancing and Imperfect Immunization

- Immunization is not static.
  - As time passes:
    - Portfolio duration declines
    - Horizon shortens
    - Interest rate movements change duration nonlinearly
- Duration matching holds only at a point in time
- Periodic rebalancing is required to:
  - Restore duration to the remaining horizon
  - Maintain immunization effectiveness

Offset Example

- Investment horizon (H): 5 years
- Bond: 6-year, 9% annual coupon, $100 par
- Initial yield (y_0): 5%
- Initial price (P_0): $120.30 per $100 par
- Macaulay duration at y_0: ≈ 5.00 (4.997)
- Strategy: Buy the bond at t=0, collect coupons for 5 years, and sell the bond at t=5 (it has 1 year remaining).

Offset Example (con.)

- Assume a parallel yield shift at t=0 to a new level y that then stays constant. We compute the future value at the horizon:

$$FV_5(y) = \sum_{t=1}^{5} C\,(1+y)^{5-t} + P_5(y)$$

- where C=9 and $P_5(y)$ is the price at t=5 of the remaining 1-year cash flow (109) :

$$P_5(y) = \frac{109}{1+y}$$

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Offset Example (con.)

- Case A: Rates fall to 4%
  - 1) Reinvestment effect (coupons reinvested to year 5)
    - $FV_{coupons} = 9[(1.04)^4+(1.04)^3+(1.04)^2+(1.04)+1]$
    - Compute each term:
      - $(1.04)^4=1.16985856$
      - $(1.04)^3=1.12486400$
      - $(1.04)^2=1.08160000$
      - $(1.04)=1.0400000011$
      - 1
      - Sum = 5.41632256
    - $FV_{coupons}= 9\times5.41632256 = 48.7469$

Offset Example (con.)

- 2) Price effect (sale price at t=5)
  - P_5(4%)=109/1.04=104.8077
  - Total horizon value FV_5(4%)= 48.7469+104.8077 = 153.5546
- Case B: Rates rise to 6%
  - Reinvestment Effect
    - $FV_{coupons} = 9[(1.06)^4+(1.06)^3+(1.06)^2+(1.06)+1]$
    - Compute each term:
      - $(1.06)^4$=1.26247696
      - $(1.06)^3$=1.19101600
      - $(1.06)^2$=1.12360000
      - (1.06)=1.06000000
      - 1
      - Sum =5.63709296
    - $FV_{coupons}$= 9×5.63709296 = 50.7338

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Offset Example (con.)

- 2) Price effect (sale price at t=5)
  - P_5(6%)=109/1.06=102.8302
  - Total horizon value FV_5(6%)= 50.7338+102.8302= 153.5640
- Compare totals:
  - $FV5$(4%)≈153.5546
  - $FV5$(6%)≈153.5640
  - They're essentially the same (difference ≈ 0.0094, < 1 bp of value), because:
    - When rates fall: sale price rises, but coupons reinvest at lower rates
    - When rates rise: sale price falls, but coupons reinvest at higher rates
  - Why not exactly identical?
    - Duration is approximately 5 (4.997, not exactly 5)
    - A 100 bp shock is not "infinitesimal," so convexity effects show up slightly

Cash Flow Matching and Dedication

- Cash Flow Matching
  - Portfolio cash flows are structured to exactly meet future liabilities
  - Eliminates reinvestment risk
  - Requires precise security selection
- Dedication:
  - Extension of cash flow matching over multiple liability dates
  - Often used by insurance companies and pension funds
- Comparison
  - Immunization: flexible, duration-based, model-dependent
  - Cash flow matching: exact, low risk, capital-intensive
  - Trade-off: Cash flow matching reduces risk but sacrifices flexibility and yield.